SEED X

You paid ₹40 for a kilo of tomatoes. How much reached the farmer?

A farmer’s harvest should not lose value before it reaches the buyer.

Better Farming Starts with Better Connections.

CodeStorm Hackathon · Agriculture, FoodTech & Rural Development

Created & Developed by Tejash Shastri and Shaurya



Live demo

Of your ₹40, the farmer may see about ₹13

You pay **₹40**

₹13 farmer | **₹27 traders · transport · storage · shop**

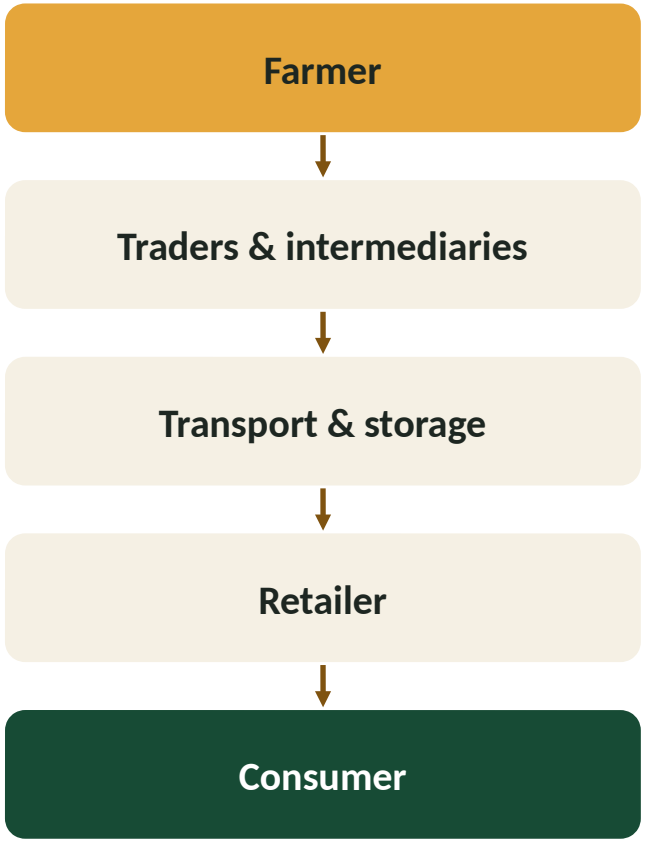
Example at the RBI-estimated tomato share (33%)

Estimated share of the consumer rupee that reaches the farmer

33% Tomato

36% Onion

37% Potato



Every hand-off takes a cut

Source: RBI Working Paper 08/2024 (3 Oct 2024), "Vegetables Inflation in India: A Study of Tomato, Onion and Potato"; also cited in a Rajya Sabha answer (29 Nov 2024). Estimates from specific studies, not a universal figure for every crop, state or sale. ₹40 → ₹13 is an illustration.

Middlemen win because they connect people

1

Dhruv can’t see the buyers

He sells to whoever comes to the village.

2

Buyers can’t find small farmers

A hostel kitchen in Bhopal needs tomatoes every week, but can’t find farms like his.

3

Three farms, three half-empty trips

Neighbours send small loads separately, so transport costs more per kilo.

4

Tomatoes don’t wait

Every day the crop sits, it loses value.



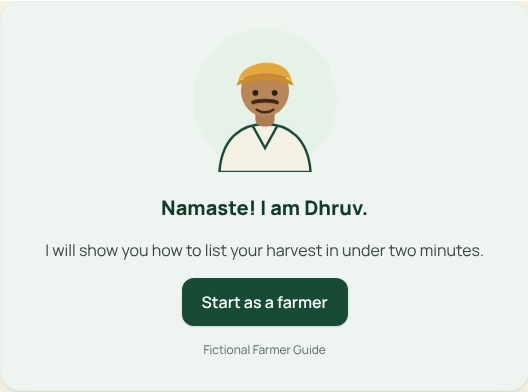
What if Dhruv’s phone did the connecting, and Dhruv kept the value?

That is SEED X.

Context: Govt. of India / NABCONS post-harvest loss study (PIB, MoFPI, 1 Aug 2025) assessed losses across 54 crops at collection, grading, packaging, transport and storage. SEED X is software; it does not build cold storage.

THE SOLUTION

Meet Dhruv: his day on SEED X



Dhruv is SEED X's farmer guide. His story here is an example.

MORNING

Lists 900 kg of tomatoes in Hindi

Sets a minimum price. His exact farm stays private.

SAME DAY

Hostel kitchen posts a need

"300 kg of tomatoes every week." Finds Dhruv by search & filters.

OFFER

Dhruv gets a notification

Accept, reject or counter with his own price, fair and on record.

DEAL

Stock reserved, order tracked

Pickup → on the way → delivered. The hostel becomes his client.

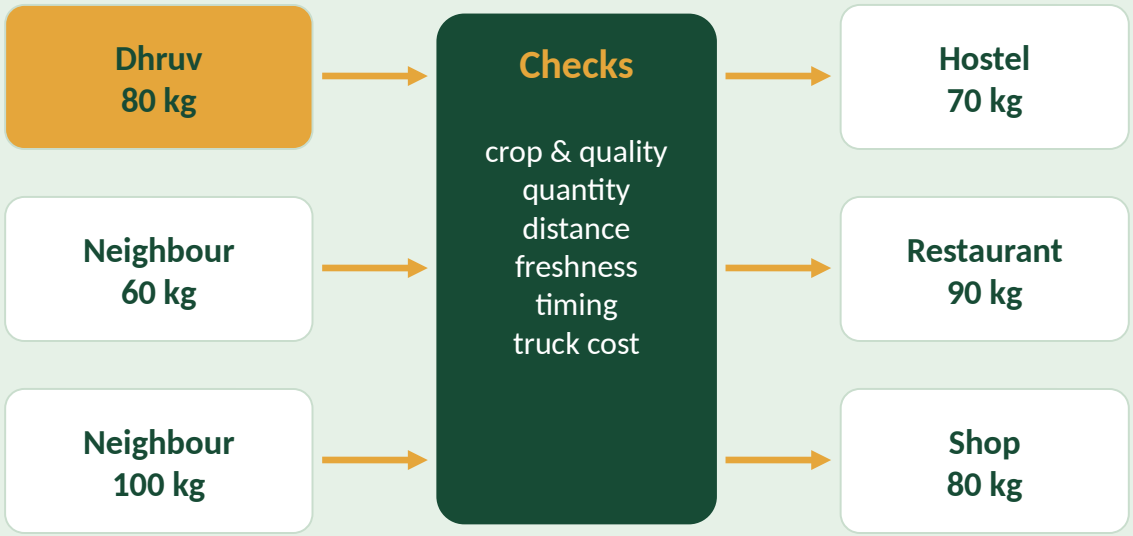
All in one app

- 46 crops
- Search & filters
- Saved searches
- Offers & counter-offers
- Notifications
- Private farm location
- Order tracking

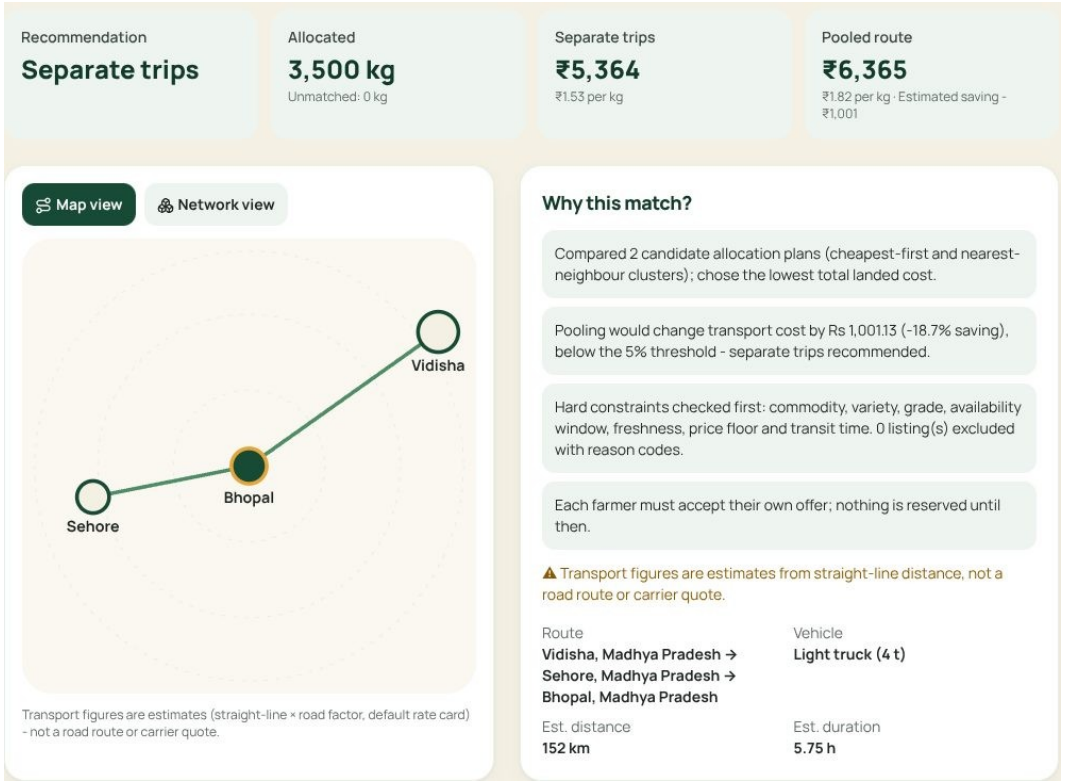
Software-only. Just the phone Dhruv already has.

The smart truck that can say “no”

Example: should Dhruv and his neighbours share one truck?



Share the truck only if it is really cheaper. If not, send separate trips, and say why.



We ran it live and it said NO. Sharing the truck would cost 18.7% more, so it recommended separate trips and explained why.

Rule-based and explainable, not AI. Transport costs are estimates (straight-line distance × road factor, standard rates), not carrier quotes. Each farmer still accepts his own offer. Live run on demo data.

A higher offer isn’t always a better deal

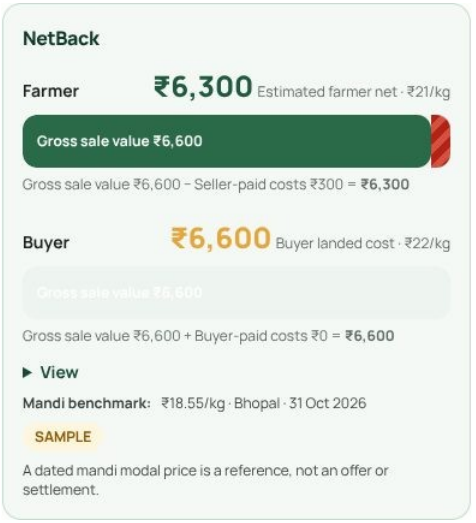
Example: what Dhruv really keeps



The hostel sees its own number

Total cost to get the food to its kitchen (landed cost), shown separately.

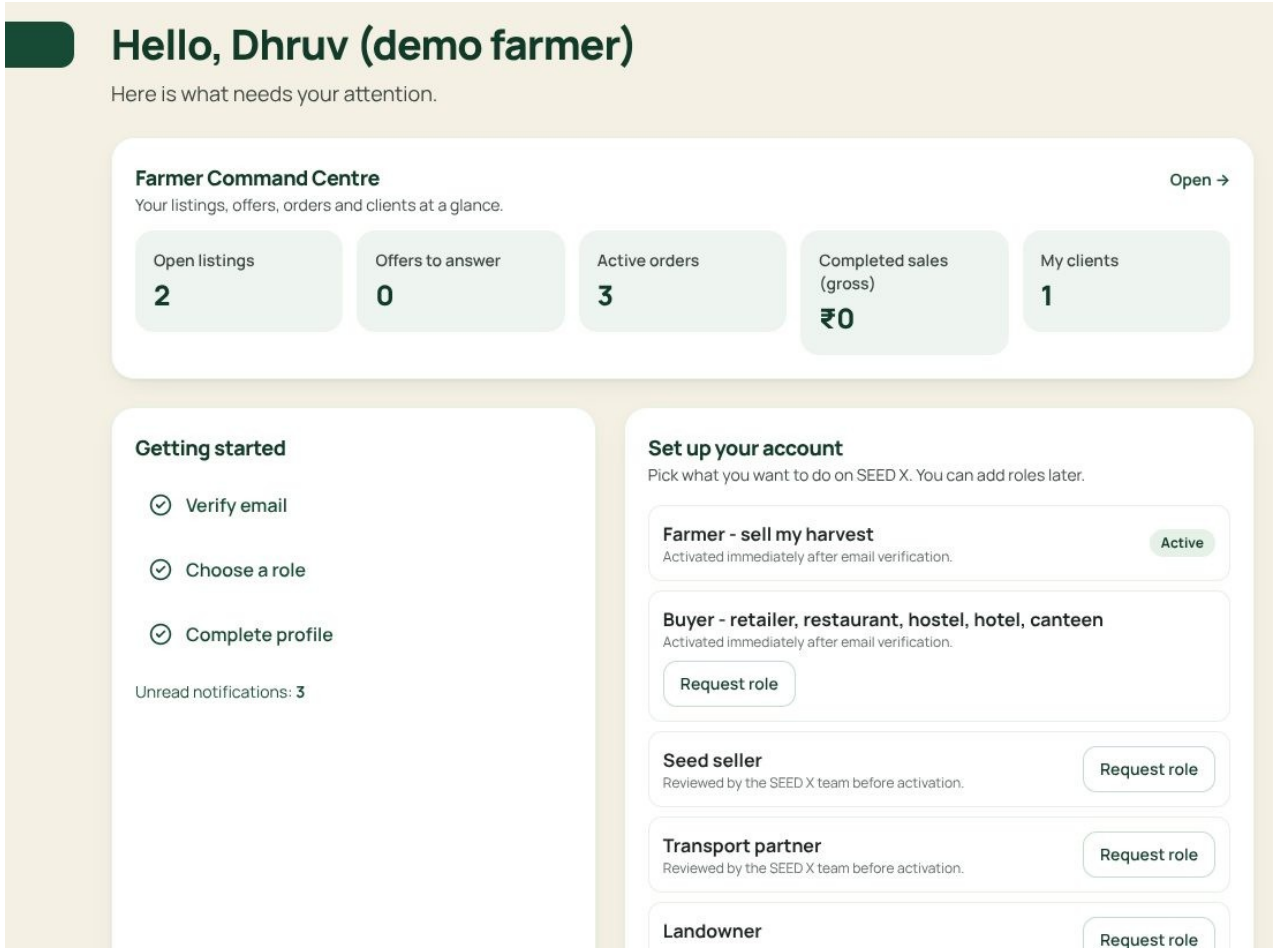
Plus a dated mandi price nearby, so Dhruv knows if the offer is fair.



Live app: 300 kg at ₹22/kg → farmer keeps ₹6,300, buyer pays ₹6,600 (demo data, estimated costs)

Example numbers. Clearer offers and less wasted cost can help farmers earn more and buyers pay fairly; shop prices are not guaranteed to fall. All money is calculated in exact paise.

Dhruv’s shop, order book and customer diary



Safe deals

Stock is reserved instantly, so no one buys the same crates twice. Track pickup → on the way → delivered.

Customers who come back

My Clients: enquiry → active → repeat client, with favourites and private notes. Buyers get My Suppliers.

A clear morning dashboard

Open listings, offers waiting, orders on the way, money from completed sales.

Help beyond selling

Seed store with honest badges, dated mandi prices, government schemes, land listings.

All shown modules work in the live app and are covered by automated tests. Screenshot: demo farmer account (synthetic data).

MADE FOR EVERY FARMER

Dhruv’s language. Dhruv’s phone

English | **हिन्दी** | বাংলা | తెలుగు | मराठी

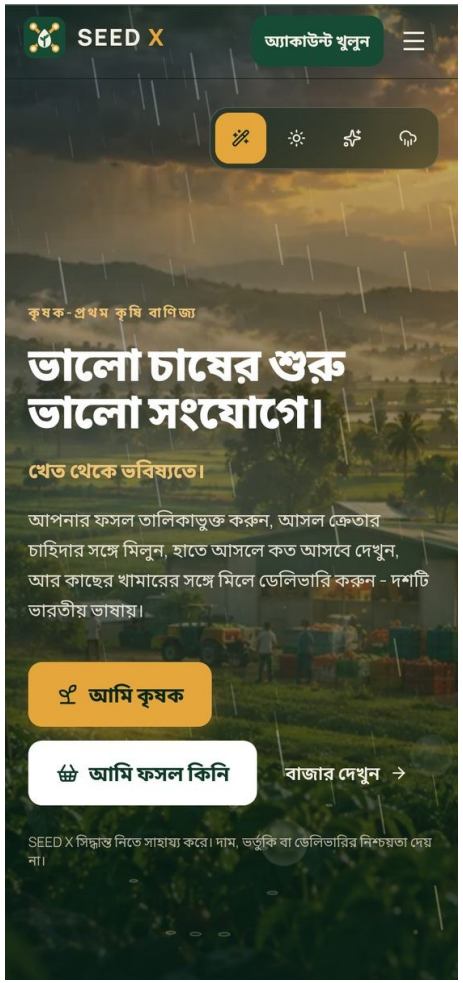
தமிழ் | ગુજરાતી | ಕನ್ನಡ | മലയാളം | ਪੰਜਾਬੀ

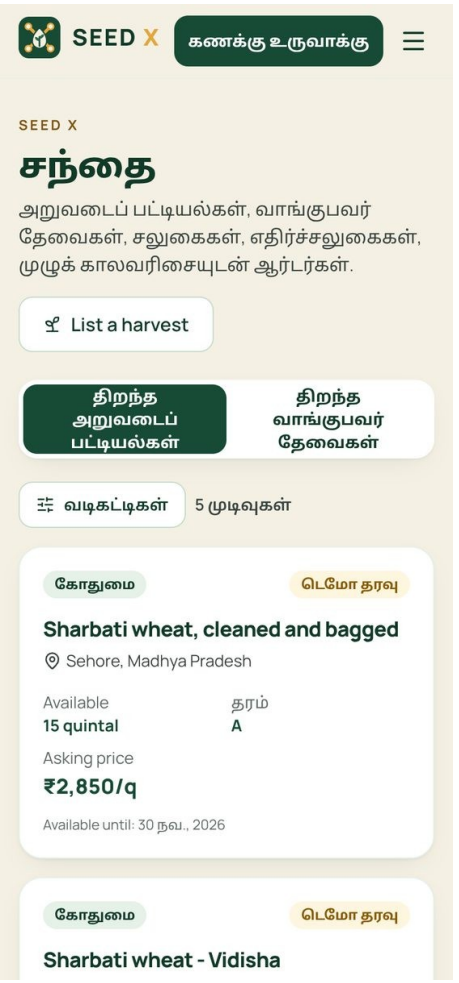
“किसान अपनी भाषा में, अपने फ़ोन पर, अपना सही दाम देख सकता है।”

Ten languages live today, crop names too. Native-speaker review in progress.

- Big buttons and short steps
- Clear kilos and quintals
- Works on an ordinary smartphone, even on slow internet
- Friendly guide: Dhruv shows new farmers how to list
- Help at first through local farmer groups and cooperatives







98% of Indian internet users already use Indian-language content.

Source: IAMAI–Kantar "Internet in India 2024": 870 million users (98%) accessed Indic-language content in 2024; all internet users, not a farmer survey. Screens: Bengali and Tamil, live app. We will test core tasks with farmers in the pilot.

Like a bank: everyone has their own locker

Your own locker

Dhruv can't open the hostel's data, and the hostel can't open his.

Address stays hidden

His exact farm unlocks only for the buyer and driver, after a real order.

Only verified traders

Everyone confirms their email before they can buy or sell.

Extra lock for admins

Admins need a second code from their phone for anything sensitive.

We attacked our own app 141 ways

✓ Another farmer reads Dhruv's private notes	BLOCKED
✓ Two buyers grab the last crate at once	ONE WINS
✓ Same order sent twice	NO DOUBLE
✓ Normal user tries an admin action	BLOCKED
✓ Upload into someone else's folder	BLOCKED

Every test passed: 141 / 141

We won't call it unhackable. We show exactly what we tested.

From the project's automated security tests (database + browser), run against the live system on 31 Oct 2026.

They built the highway. We build Dhruv's road

THE HIGHWAY · already exists

e-NAM

Online trading between mandis, with price discovery and auctions.

AGMARKNET

Daily mandi prices for the whole country.

THE ROAD · SEED X

- Dhruv's own crops and stock
- Nearby direct buyers: hostels, restaurants, shops
- Shared trucks only when cheaper
- His real take-home money
- Regular customers who come back
- Uses public mandi prices as a reference

Not a mock-up: it's live

Modern web app

Fast on any phone

Secure cloud database

Locked per user

Hosted on Cloudflare

Worldwide, auto-updates

Live today

Scan the QR and try it

Sources: enam.gov.in; data.gov.in / AGMARKNET daily mandi prices. SEED X complements public market systems; it does not claim to be the first digital mandi. Entirely software-based.

SEE IT LIVE · THE PLAN

See it live, in 60 seconds

1. Home page: a farmer's world
2. Dhruv's tomato listing
3. NetBack: what he really keeps
4. ClusterCart: share the truck or not?
5. My Suppliers & morning dashboard
6. Switch to हिन्दी

Live: seedxsss.tejash-shastri-rx.workers.dev
Backup video ready (79 s)

How SEED X earns

Farmers list free. Always.

Small fee on completed orders

paid by buyers

ClusterCart planning

for hostels, canteens, retailers

Start small

one district · one crop · one farmer group

Proposed, to be tested in the pilot

What we'll measure

- More money per kilo for the farmer
- Lower cost per kilo for buyers
- Transport cost per kilo
- Time to find a buyer
- On-time deliveries
- Buyers who come back
- Less crop wasted

Today, of your ₹40, maybe ₹13 reaches Dhruv. He doesn't need sympathy. He needs to see his buyers, his costs and his real price.